Project Name: Student Performance Analytics & Decision Support System
Client: Mega Central High School
Project Sponsor: Chairman, Mega Central High School
Business Analyst: Ramyashree GV
Project Manager: Phanikrishna
Project Duration: 22 June 2026 – 03 August 2026 (6 weeks)

Project Background:

Mega Central High School conducts periodic offline classroom assessments for Grade 10 students. After examinations are evaluated, subject teachers record student marks and question-wise performance in individual Excel spreadsheets. These spreadsheets are later consolidated for academic reporting.

Because each department maintains data independently, preparing consolidated reports is time-consuming, inconsistent, and prone to manual errors.

The school intends to implement a centralized Student Performance Analytics & Decision Support System that integrates assessment data from departmental spreadsheets, automates analytics, and provides interactive dashboards for teachers and management.

Business Problem:

The school currently experiences several operational challenges:

- Student performance data is stored in multiple Excel files.
- Academic reports are prepared manually.
- Weak students are identified only after final examinations.
- Chapter-level performance is not monitored.
- Teachers spend significant time consolidating results.
- School management lacks real-time visibility into academic performance.

Project Objective:

Develop a centralized analytics system that enables teachers and school management to monitor academic performance, identify at-risk students, analyze subject and chapter performance, and support timely academic interventions through interactive dashboards and automated reporting.

Project Scope:

In Scope:

- Student master data management
- Question bank management

- Test management
- Digitized question-wise assessment record management
- Automated score calculation
- Student progress monitoring
- Risk identification
- Question analytics
- Chapter analytics
- Executive dashboards
- Power BI reporting

Out of Scope:

- Online examination platform
- Student login portal
- Parent mobile application
- Fee management
- Attendance management
- Learning Management System (LMS)
- AI-based question generation

Project Goals:

- Reduce manual reporting effort.
- Enable data-driven academic decisions.
- Identify high-risk students earlier.
- Improve chapter-level teaching effectiveness.
- Provide centralized performance reporting.

Success Criteria:

KPI	Target
Dashboard availability	100%
Report preparation time	Reduce by 80%
Manual score calculation	Eliminated
Student performance visibility	Real-time
High-risk student identification	Before final exams
Dashboard adoption by teachers	>90%

Constraints:

- Limited historical data available.
- Initial implementation limited to Grade 10.
- Only four core subjects included.
- Data refresh performed manually.
- Internet connectivity may be limited during report access.
- Historical data limited to the current academic year.

Assumptions:

- Teachers enter assessment data accurately.
- All students participate in scheduled assessments.
- School management will use dashboard insights for academic planning.
- Student identifiers remain unique across datasets.

Risks:

Risk	Impact	Mitigation
Incorrect assessment data	High	Data validation rules
Missing student responses	Medium	Data quality monitoring
User adoption challenges	Medium	Training & user manuals
Scope changes	Medium	Formal change request process
Data privacy concerns	High	Role-based dashboard access

Key Deliverables:

- Project Charter
- Business Requirements Document (BRD)
- Functional Requirements Document (FRD)
- User Stories
- Process Flow Diagrams
- Data Dictionary
- SQL Scripts
- Python ETL Pipeline
- Power BI Dashboard
- UAT Documentation
- Final Project Presentation